

- last few question of quiz
 - poles important
- tut also
 - degeneration
 - how it affects the GM and R
 - all the circuit derivation

- second part
 - diff pair analysis
 - split half with virtual ground
 - the DM input definition must get right
 - operating point for saturation, range of CM input
 - how CM gain appear
 - D to S
 - the two paths
 - how the poles and zero appear
 - no KCL KVL stuff
 - assoc with poles also
 - op amp
 - how the gain stack
 - DC biasing point
 - how the mirror constraints everything
- the frequency response standard formula
 - with the $1 - X$ and $1 + X$
- Zero is always missed by Miller
 - need to catch with the signal path with series Cap (feed forward)
 - get value by making $v_{out} = 0$